Analysis

Cam Smithers

2026-06-26

Analysis

Packages

```
set.seed(157)
library(tidyverse)
library(glue)
library(zoo)
library(psych)
library(ggridges)
library(viridis)
library(hrbrthemes)
```

Functions Script

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
source('inR/Functions.R')
```

Analysis Cleaning

Data Sets

Loading in the data sets that I'll need for analysis.

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
team_box_scores <- readRDS(
    'Data-NBA/Current/team_box_scores_current.rds')
player_box_scores <- readRDS(
    'Data-NBA/Current/full_box_player_current.rds')
team_season_stats <- readRDS(
    'Data-NBA/Current/team_season_stats_current.rds')
player_season_stats <- readRDS(
    'Data-NBA/Current/player_season_stats_current.rds')
```

Applying a function I wrote that fixes some aesthetic issues with player names. This will occur with foreign players that have special characters or foreign characters in their name.

```
player_box_scores <- fixplayername(player_box_scores)
player_season_stats <- fixplayername(player_season_stats)
```

Outcomes and Small Metrics

Outcome I'm adding the points from the opposing team so I can create an outcome variable based on the margin of the outcome. This will be multinomial regression with four different classifications. Those are...

```
opponent_stats <- team_box_scores %>%
    select(gamedate, team, pts) %>%
    rename('opp_pts'='pts')

team_box_scores <- team_box_scores %>%
    left_join(opponent_stats, by = c('gamedate', 'opponent'='team'))
```

```
team_box_scores <- team_box_scores %>%
    mutate(
        ptsdiff = pts - opp_pts,
        twoleveloc = if_else(offrating > defrating, 1, 0),
        threeleveloc = case_when(
            ptsdiff >= 11 ~ 2,
            ptsdiff %in% 1:10 ~ 1,
            TRUE ~ 0),
        fourleveloc = case_when(
            ptsdiff <= -11 ~ 0,
            ptsdiff %in% -1:-10 ~ 1,
            ptsdiff %in% 1:10 ~ 2,
            ptsdiff >= 11 ~ 3)) %>%
    mutate(
        twoleveloc = factor(twoleveloc,
                            levels = c(0, 1),
                            labels = c('Loss', 'Win')),
        threeleveloc = factor(threeleveloc,
                              levels = c(0, 1, 2),
                              labels = c('Loss', 'Basic Win', 'Blowout Win')),
        fourleveloc = factor(fourleveloc,
                             levels = c(0, 1, 2, 3),
                             labels = c('Blowout Loss', 'Basic Loss',
                                        'Basic Win', 'Blowout Win')))
```

Conditional Outcomes I want to look at how missing a player affects the chance of a team to win a game; however, I have to determine which players are important for the outcome of the game. I'm going to look at few metrics and use those to determine a player's significance. For the different metrics, different players may end up being the most important player. Those metrics include...

1. Win Shares: `ws`
2. Value Over Replacement Player (VORP): `vorp`
3. Player Efficiency Rating: `player_effrtg`

I'm also going to create a variable that has a unique id for each best player by a given metric.

```
bp_by_ws <- best_player_metric(player_season_stats, ws, 20, 10) %>%
    rename('ws_name'='name') %>%
    unite('ws_team_id', c('year', 'team', 'ws_player_id'),
          sep = '_', remove = FALSE) %>%
    mutate(ws_team_id = paste0('ws_', ws_team_id)) %>%
    relocate(ws_team_id, .after = last_col())
bp_by_vorp <- best_player_metric(player_season_stats, vorp, 20, 10) %>%
    rename('vorp_name'='name') %>%
```

```
    unite('vorp_team_id', c('year', 'team', 'vorp_player_id'),
          sep = '_', remove = FALSE) %>%
    mutate(vorp_team_id = paste0('vorp_', vorp_team_id)) %>%
    relocate(vorp_team_id, .after = last_col())
bp_by_per <- best_player_metric(player_season_stats, player_effrtg, 20, 10) %>%
    rename('per_name'='name') %>%
    unite('per_team_id', c('year', 'team', 'player_effrtg_player_id'),
          sep = '_', remove = FALSE) %>%
    mutate(per_team_id = paste0('per_', per_team_id)) %>%
    relocate(per_team_id, .after = last_col())
```

Here I'm going to create a variable within each data set that will allow to check for player availability when coding a future outcome variable

Joining the data from the best player by a certain metric data sets to later determine if player was on the court and how that affected the outcome.

```
team_box_scores_2 <- team_box_scores %>%
    left_join(bp_by_ws, by = c('team', 'season'='year')) %>%
    left_join(bp_by_vorp, by = c('team', 'season'='year')) %>%
    left_join(bp_by_per, by = c('team', 'season'='year')) %>%
    mutate(
        ws_tm_game_id = paste0(ws_team_id, '_', gamedate),
        vorp_tm_game_id = paste0(vorp_team_id, '_', gamedate),
        per_tm_game_id = paste0(per_team_id, '_', gamedate))

player_box_scores_2 <- player_box_scores %>%
    left_join(bp_by_ws, by = c('team', 'season'='year', 'name'='ws_name')) %>%
    left_join(bp_by_vorp, by = c('team', 'season'='year', 'name'='vorp_name')) %>%
    left_join(bp_by_per, by = c('team', 'season'='year', 'name'='per_name')) %>%
    mutate(
        ws_plyr_game_id = if_else(
            !is.na(ws_team_id), paste0(ws_team_id, '_', gamedate), NA),
        vorp_plyr_game_id = if_else(
            !is.na(vorp_team_id), paste0(vorp_team_id, '_', gamedate), NA),
        per_plyr_game_id = if_else(
            !is.na(per_team_id), paste0(per_team_id, '_', gamedate), NA))
```

After creating the data sets necessary to determine whether a team's best player (by a given metric) was available I can now create an outcome variable to check for it. I'm also going to add a variable to determines whether a game was a regular season game or playoff game.

Note: *I know I don't have to use multiple mutate functions; however, at times I like to use multiple as it helps me organize my code as I'm writing it.*

```
team_box_scores_3 <- team_box_scores_2 %>%
    mutate(
        ws_bpa = if_else(ws_tm_game_id %in%
                             player_box_scores_2$ws_plyr_game_id, 1, 0),
        vorp_bpa = if_else(vorp_tm_game_id %in%
                            player_box_scores_2$vorp_plyr_game_id, 1, 0),
        per_bpa = if_else(per_tm_game_id %in%
                           player_box_scores_2$per_plyr_game_id, 1, 0)) %>%
    mutate(
        ws_bpa_multi = case_when(
            twoleveloc == 'Loss' & ws_bpa == 0 ~ 0,
```

```
            twoleveloc == 'Loss' & ws_bpa == 1 ~ 1,
            twoleveloc == 'Win' & ws_bpa == 0 ~ 2,
            twoleveloc == 'Win' & ws_bpa == 1 ~ 3),
        vorp_bpa_multi = case_when(
            twoleveloc == 'Loss' & vorp_bpa == 0 ~ 0,
            twoleveloc == 'Loss' & vorp_bpa == 1 ~ 1,
            twoleveloc == 'Win' & vorp_bpa == 0 ~ 2,
            twoleveloc == 'Win' & vorp_bpa == 1 ~ 3),
        per_bpa_multi = case_when(
            twoleveloc == 'Loss' & per_bpa == 0 ~ 0,
            twoleveloc == 'Loss' & per_bpa == 1 ~ 1,
            twoleveloc == 'Win' & per_bpa == 0 ~ 2,
            twoleveloc == 'Win' & per_bpa == 1 ~ 3)) %>%
    mutate(
        ws_bpa_multi = factor(ws_bpa_multi,
                              levels = c(0, 1, 2, 3),
                              labels = c('Loss Without', 'Loss With',
                                         'Win Without', 'Win With')),
        vorp_bpa_multi = factor(vorp_bpa_multi,
                                levels = c(0, 1, 2, 3),
                                labels = c('Loss Without', 'Loss With',
                                           'Win Without', 'Win With')),
        per_bpa_multi = factor(per_bpa_multi,
                               levels = c(0, 1, 2, 3),
                               labels = c('Loss Without', 'Loss With',
                                          'Win Without', 'Win With'))) %>%
    mutate(
        gametype = case_when(
            #2021 Season
            (gamedate >= as.Date('2021-05-22') &
                 gamedate <= as.Date('2021-07-20')) |
                #2022 Season
                (gamedate >= as.Date('2022-04-16') &
                     gamedate <= as.Date('2022-06-16')) |
                #2023 Season
                (gamedate >= as.Date('2023-04-15') &
                     gamedate <= as.Date('2023-06-12')) |
                #2024 Season
                (gamedate >= as.Date('2024-04-20') &
                     gamedate<= as.Date('2024-06-17')) |
                #2025 Season
                (gamedate >= as.Date('2025-04-19') &
                     gamedate <= as.Date('2025-06-2022')) |
                #2026 Season
                 (gamedate >= as.Date('2026-04-18') &
                     gamedate <= as.Date('2026-06-13')) ~ 'Playoff Game',
            TRUE ~ 'Regular Season Game'))
```

Here I'm going to create some variables that will be used as possible predictors when testing regression models. There are great variables to work with in the data set so far; however, I want to create some rolling averages so I can measure how teams are doing across seasons and within a certain window.

```
regsnbox <- team_box_scores_3 %>%
    filter(gametype == 'Regular Season Game') %>%
```

```
    group_by(team) %>%
    arrange(desc(gamedate), .by_group=TRUE) %>%
    mutate(
        roll_fg_pct = rollmean(fg_pct, k=10, fill=NA, align='left'),
        roll_threefg_pct = rollmean(threefg_pct, k=10, fill=NA, align='left'),
        roll_pts = rollmean(pts, k=10, fill=NA, align='left'),
        roll_three_par = rollmean(three_par, k=10, fill=NA, align='left'),
        roll_dreb_pct = rollmean(dreb_pct, k=10, fill=NA, align='left'),
        roll_ftar = rollmean(ftar, k=10, fill=NA, align='left'),
        roll_ast_pct = rollmean(ast_pct, k=10, fill=NA, align='left'),
        roll_offrating = rollmean(offrating, k=10, fill=NA, align='left'),
        roll_defrating = rollmean(defrating, k=10, fill=NA, align='left'),
        roll_opp_pts = rollmean(opp_pts, k=10, fill=NA, align='left'),
        roll_ptsdiff = rollmean(ptsdiff, k=10, fill=NA, align='left')) %>%
    select(gamedate, team, starts_with('roll_'))
pstsnbox <- team_box_scores_3 %>%
    filter(gametype == 'Playoff Game') %>%
    group_by(team) %>%
    arrange(desc(gamedate), .by_group=TRUE) %>%
    mutate(
        roll_fg_pct = rollmean(fg_pct, k=3, fill=NA, align='left'),
        roll_threefg_pct = rollmean(threefg_pct, k=10, fill=NA, align='left'),
        roll_pts = rollmean(pts, k=3, fill=NA, align='left'),
        roll_three_par = rollmean(three_par, k=3, fill=NA, align='left'),
        roll_dreb_pct = rollmean(dreb_pct, k=3, fill=NA, align='left'),
        roll_ftar = rollmean(ftar, k=3, fill=NA, align='left'),
        roll_ast_pct = rollmean(ast_pct, k=3, fill=NA, align='left'),
        roll_offrating = rollmean(offrating, k=3, fill=NA, align='left'),
        roll_defrating = rollmean(defrating, k=3, fill=NA, align='left'),
        roll_opp_pts = rollmean(opp_pts, k=3, fill=NA, align='left'),
        roll_ptsdiff = rollmean(ptsdiff, k=3, fill=NA, align='left')) %>%
    select(gamedate, team, starts_with('roll_'))
fullsnbox <- rbind(regsnbox, pstsnbox)
```

Using the rounding function across across various columns to clean up the rolling averages.

```
team_box_scores_4 <- team_box_scores_3 %>%
    left_join(fullsnbox, by = c('gamedate', 'team')) %>%
    mutate(across(c('roll_fg_pct','roll_threefg_pct','roll_three_par',
                    'roll_ftar'), round, 3)) %>%
    mutate(across(c('roll_pts','roll_dreb_pct','roll_ast_pct',
                    'roll_offrating','roll_defrating','roll_opp_pts',
                    'roll_ptsdiff'), round, 1)) %>%
    select(-c(ws_name, ws, ws_player_id, ws_team_id, vorp_name, vorp,
              per_name, player_effrtg, player_effrtg_player_id,
              per_team_id, ws_tm_game_id, vorp_tm_game_id,
              per_tm_game_id, vorp_team_id, vorp_player_id))
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(...)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
```

```
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

Summary Data

Adding a non-moving average for the data set, in the off chance I may want to use it for something.

```
team_box_summary <- team_box_scores_4 %>%
    group_by(season, team) %>%
    summarise(
        avg_fg_pct = round(mean(fg_pct, na.rm=TRUE), 3),
        avg_3fg_pct = round(mean(threefg_pct, na.rm=TRUE), 3),
        avg_pts = round(mean(pts, na.rm=TRUE), 1),
        avg_offrating = round(mean(offrating, na.rm=TRUE), 1),
        avg_defrating = round(mean(defrating, na.rm=TRUE), 1))
```

```
## `summarise()` has grouped output by 'season'. You can override using the
## `.groups` argument.
```

Data Export for Visualizations

Export data as RDS files for visualizations in R.

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA/Data-NBA/Current')
saveRDS(team_box_scores_4, file = 'teamvisdata.rds')
saveRDS(team_box_summary, file = 'teamboxsummary.rds')
```

Export data as CSV files for Python dashboard.

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA/dashboard')
write_csv(team_box_scores_4, file='teamvisdata.csv')
write_csv(team_box_summary, file='teamboxsummary.csv')
```

Basic Data Dive

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
teamvisdata <- readRDS('Data-NBA/Current/teamvisdata.rds')
teamboxsummary <- readRDS('Data-NBA/Current/teamboxsummary.rds')
```

Here I'm using the `describe` function from the psych package to generate a table with summary statistics.

1. N: Number of Observations
2. Mean: Average of the Obversations
3. SD: The Standard Deviation
4. Median: The Number in the Middle of the Observations
5. Trimmed: Mean Calculated Without Extreme Values
6. MAD: Mean Absolute Deviation
7. Min: Minimum Value
8. Max: Maximum Value
9. Range: Difference Between Max and Min
10. Skew: Symmetry of the Data Set
11. Kurtosis: Determine the Frequency of Outliers

12. SE: Standard Error

```
describe(teamvisdata)
```

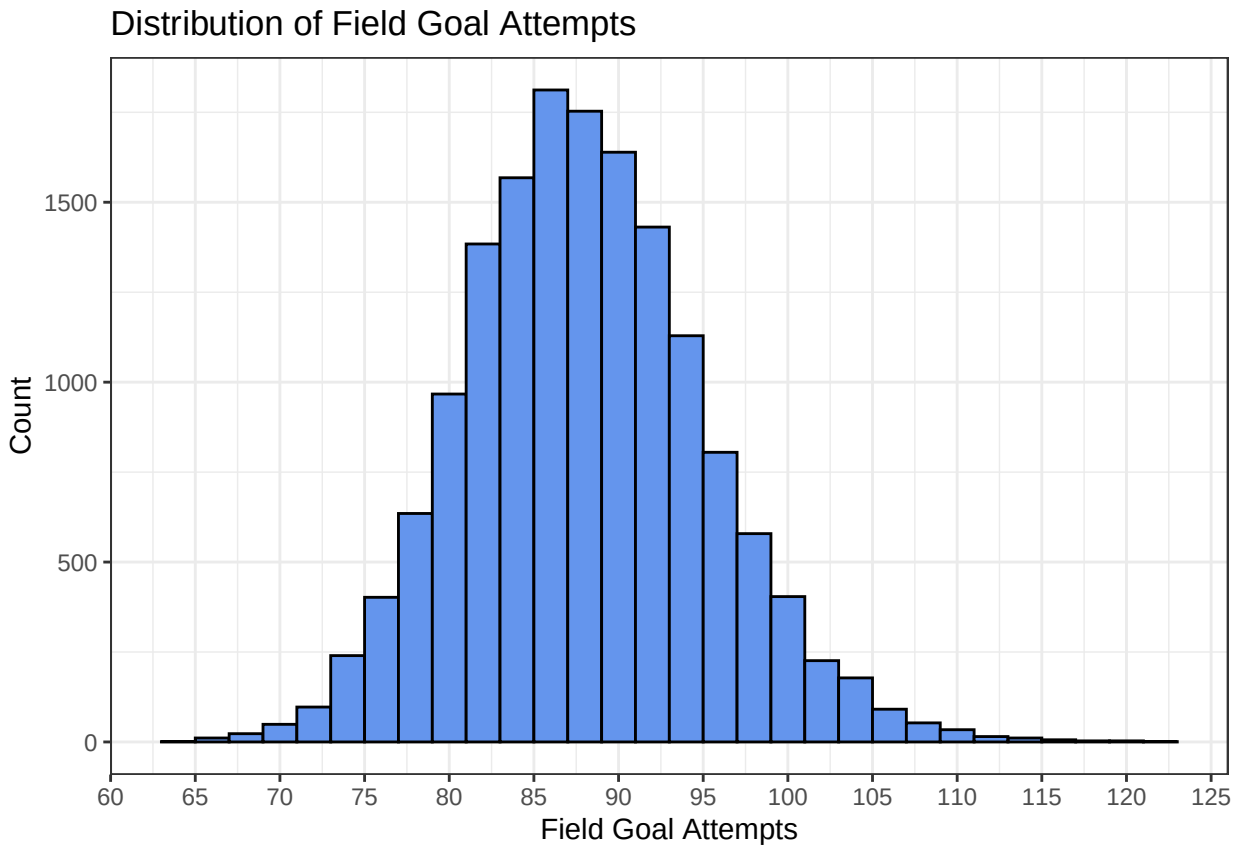
```
##                  vars     n    mean    sd  median trimmed   mad     min     max
## gamedate            1 15550     NaN    NA      NA     NaN    NA     Inf    -Inf
## team                2 15550   15.41  8.61   15.00   15.40 10.38    1.00   30.00
## opponent            3 15550   15.41  8.61   15.00   15.40 10.38    1.00   30.00
## min                 4 15550  241.41  6.41  240.00  240.00  0.00  240.00  315.00
## fg                  5 15550   41.45  5.31   41.00   41.37  5.93   22.00   65.00
## fga                 6 15550   88.47  7.11   88.00   88.30  7.41   64.00  122.00
## fg_pct              7 15550    0.47  0.06    0.47    0.47  0.06    0.28    0.69
## threefg             8 15550   12.81  3.93   13.00   12.69  4.45    2.00   29.00
## threefga            9 15550   35.55  6.99   35.00   35.40  7.41   10.00   65.00
## threefg_pct        10 15550    0.36  0.08    0.36    0.36  0.08    0.07    0.68
## ft                 11 15550   17.46  5.88   17.00   17.25  5.93    0.00   47.00
## fta                12 15550   22.38  7.05   22.00   22.14  7.41    0.00   59.00
## ft_pct             13 15549    0.78  0.10    0.79    0.78  0.10    0.30    1.00
## oreb               14 15550   10.61  3.89   10.00   10.43  4.45    0.00   29.00
## dreb               15 15550   33.24  5.42   33.00   33.16  5.93   14.00   60.00
## treb               16 15550   43.84  6.67   44.00   43.72  7.41   20.00   74.00
## ast                17 15550   25.62  5.19   25.00   25.49  4.45    8.00   50.00
## stl                18 15550    7.73  2.98    7.00    7.60  2.97    0.00   22.00
## blk                19 15550    4.85  2.47    5.00    4.70  2.97    0.00   19.00
## tov                20 15550   13.97  3.93   14.00   13.86  4.45    1.00   31.00
## pf                 21 15550   19.44  4.16   19.00   19.36  4.45    4.00   37.00
## pts                22 15550  113.16 12.76  113.00  113.05 13.34   66.00  176.00
## ts_pct             23 15550    0.58  0.06    0.58    0.58  0.06    0.36    0.82
## efg_pct            24 15550    0.54  0.07    0.54    0.54  0.07    0.30    0.81
## three_par          25 15550    0.40  0.07    0.40    0.40  0.08    0.11    0.72
## ftar               26 15550    0.26  0.09    0.25    0.25  0.09    0.00    0.70
## oreb_pct           27 15550   24.01  7.56   23.80   23.86  7.56    0.00   58.80
## dreb_pct           28 15550   75.99  7.56   76.20   76.14  7.56   41.20  100.00
## treb_pct           29 15550   50.00  5.47   50.00   50.00  5.49   29.00   71.00
## ast_pct            30 15550   61.77  9.44   61.90   61.88  9.49   25.70   96.90
## stl_pct            31 15550    7.77  2.92    7.60    7.66  2.97    0.00   21.10
## blk_pct            32 15550    9.16  4.50    8.80    8.93  4.45    0.00   40.40
## tov_pct            33 15550   12.43  3.39   12.30   12.35  3.41    1.00   26.70
## usage_rate         34 15550  100.00  0.00  100.00  100.00  0.00  100.00  100.00
## offrating          35 15550  114.14 11.84  114.20  114.16 12.01   68.30  163.40
## defrating          36 15550  114.14 11.84  114.20  114.16 12.01   68.30  163.40
## homeaway           37 15550    1.50  0.50    1.50    1.50  0.74    1.00    2.00
## season             38 15550 2023.55  1.69 2024.00 2023.56  1.48 2021.00 2026.00
## opp_pts            39 15550  113.16 12.76  113.00  113.05 13.34   66.00  176.00
## ptsdiff            40 15550    0.00 15.54    0.00    0.00 14.83  -73.00   73.00
## twoleveloc         41 15550    1.50  0.50    1.50    1.50  0.74    1.00    2.00
## threeleveloc       42 15550    1.74  0.82    1.50    1.68  0.74    1.00    3.00
## fourleveloc        43 15550    2.50  1.10    2.50    2.50  0.74    1.00    4.00
## ws_bpa             44 15550    0.87  0.34    1.00    0.96  0.00    0.00    1.00
## vorp_bpa           45 15550    0.83  0.38    1.00    0.91  0.00    0.00    1.00
## per_bpa            46 15550    0.74  0.44    1.00    0.80  0.00    0.00    1.00
## ws_bpa_multi       47 15550    2.87  1.08    2.50    2.94  2.22    1.00    4.00
## vorp_bpa_multi     48 15550    2.83  1.11    2.50    2.91  2.22    1.00    4.00
## per_bpa_multi      49 15550    2.74  1.14    2.50    2.80  2.22    1.00    4.00
## gametype           50 15550    1.93  0.25    2.00    2.00  0.00    1.00    2.00
```

```
## roll_fg_pct        51 15222    0.47  0.02    0.47    0.47  0.02    0.35    0.60
## roll_threefg_pct   52 15029    0.36  0.03    0.36    0.36  0.03    0.25    0.47
## roll_pts           53 15222  113.17  6.10  113.30  113.29  6.08   85.30  134.30
## roll_three_par     54 15222    0.40  0.05    0.40    0.40  0.05    0.24    0.58
## roll_dreb_pct      55 15222   75.99  3.10   76.10   76.05  2.97   60.00   91.40
## roll_ftar          56 15222    0.26  0.04    0.25    0.25  0.04    0.08    0.49
## roll_ast_pct       57 15222   61.79  4.95   61.70   61.80  4.89   38.40   77.80
## roll_offrating     58 15222  114.18  5.44  114.30  114.23  5.49   86.30  139.70
## roll_defrating     59 15222  114.15  5.19  114.10  114.14  5.04   91.30  136.40
## roll_opp_pts       60 15222  113.15  6.28  113.10  113.15  6.38   85.30  136.50
## roll_ptsdiff       61 15222    0.03  7.20    0.30    0.13  7.12  -36.70   39.70
##                   range  skew kurtosis   se
## gamedate           -Inf    NA       NA   NA
## team              29.00  0.00    -1.19 0.07
## opponent          29.00  0.00    -1.19 0.07
## min               75.00  5.00    27.87 0.05
## fg                43.00  0.14     0.01 0.04
## fga               58.00  0.29     0.36 0.06
## fg_pct             0.41  0.10    -0.06 0.00
## threefg           27.00  0.34     0.09 0.03
## threefga          55.00  0.22     0.04 0.06
## threefg_pct        0.62  0.08     0.03 0.00
## ft                47.00  0.38     0.16 0.05
## fta               59.00  0.37     0.20 0.06
## ft_pct             0.70 -0.40     0.41 0.00
## oreb              29.00  0.48     0.33 0.03
## dreb              46.00  0.16     0.05 0.04
## treb              54.00  0.21     0.12 0.05
## ast               42.00  0.26     0.11 0.04
## stl               22.00  0.44     0.18 0.02
## blk               19.00  0.64     0.64 0.02
## tov               30.00  0.28     0.04 0.03
## pf                33.00  0.19     0.14 0.03
## pts              110.00  0.10     0.01 0.10
## ts_pct             0.47  0.11    -0.06 0.00
## efg_pct            0.51  0.15    -0.03 0.00
## three_par          0.62  0.11    -0.02 0.00
## ftar               0.70  0.55     0.48 0.00
## oreb_pct          58.80  0.21    -0.01 0.06
## dreb_pct          58.80 -0.21    -0.01 0.06
## treb_pct          42.00  0.00     0.00 0.04
## ast_pct           71.20 -0.12    -0.08 0.08
## stl_pct           21.10  0.39     0.10 0.02
## blk_pct           40.40  0.59     0.66 0.04
## tov_pct           25.70  0.25     0.01 0.03
## usage_rate         0.00   NaN      NaN 0.00
## offrating         95.10 -0.01    -0.04 0.09
## defrating         95.10 -0.01    -0.04 0.09
## homeaway           1.00  0.00    -2.00 0.00
## season             5.00 -0.02    -1.25 0.01
## opp_pts          110.00  0.10     0.01 0.10
## ptsdiff          146.00  0.00     0.31 0.12
## twoleveloc         1.00  0.00    -2.00 0.00
## threeleveloc       2.00  0.51    -1.34 0.01
```

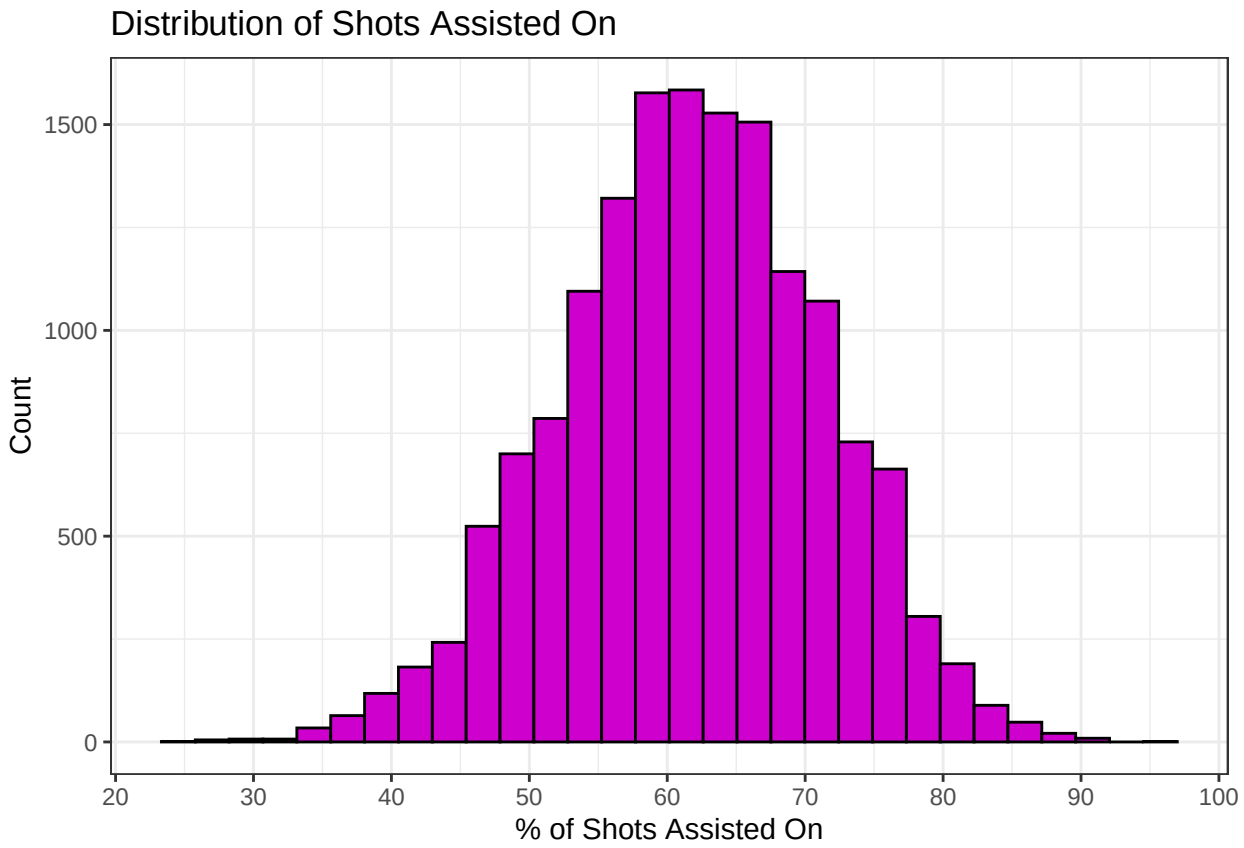
```
## fourleveloc        3.00  0.00    -1.33 0.01
## ws_bpa             1.00 -2.17     2.69 0.00
## vorp_bpa           1.00 -1.75     1.06 0.00
## per_bpa            1.00 -1.08    -0.84 0.00
## ws_bpa_multi       3.00 -0.12    -1.58 0.01
## vorp_bpa_multi     3.00 -0.13    -1.54 0.01
## per_bpa_multi      3.00 -0.11    -1.50 0.01
## gametype           1.00 -3.53    10.43 0.00
## roll_fg_pct        0.24 -0.04     0.28 0.00
## roll_threefg_pct   0.22  0.03     0.03 0.00
## roll_pts          49.00 -0.20     0.07 0.05
## roll_three_par     0.34  0.19    -0.08 0.00
## roll_dreb_pct     31.40 -0.21     0.62 0.03
## roll_ftar          0.41  0.41     0.82 0.00
## roll_ast_pct      39.40 -0.07     0.16 0.04
## roll_offrating    53.40 -0.10     0.07 0.04
## roll_defrating    45.10 -0.03     0.25 0.04
## roll_opp_pts      51.20 -0.04     0.11 0.05
## roll_ptsdiff      76.40 -0.13     0.15 0.06
```

Now let's create some basic plots to get a better understanding of our data set.

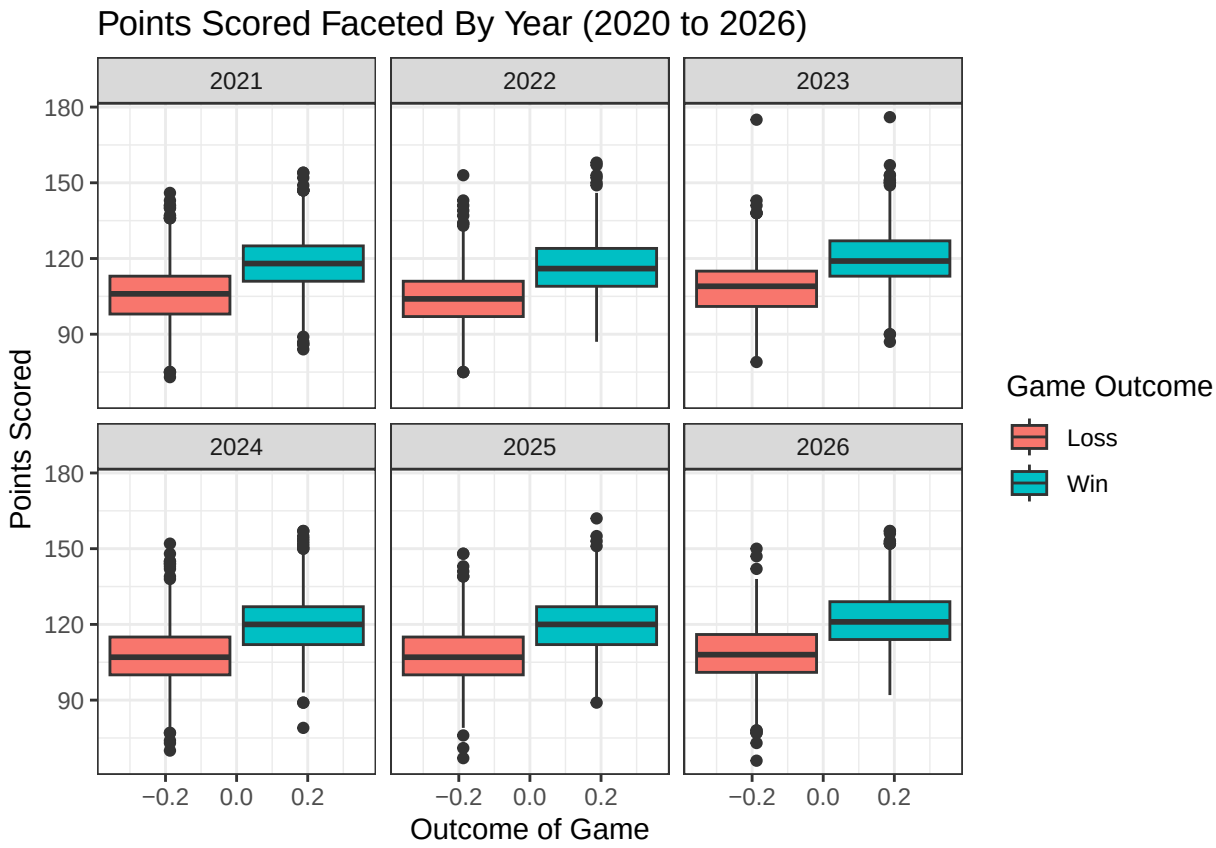
```
hist1plot <- ggplot(data=teamvisdata,
                    aes(x=fga)) +
    geom_histogram(bins=30, color='black', fill='cornflowerblue') +
    labs(
        title='Distribution of Field Goal Attempts',
        x='Field Goal Attempts',
        y='Count') +
    theme_bw() +
    scale_x_continuous(breaks = seq(from = 60, to = 130, by = 5))
hist1plot
```



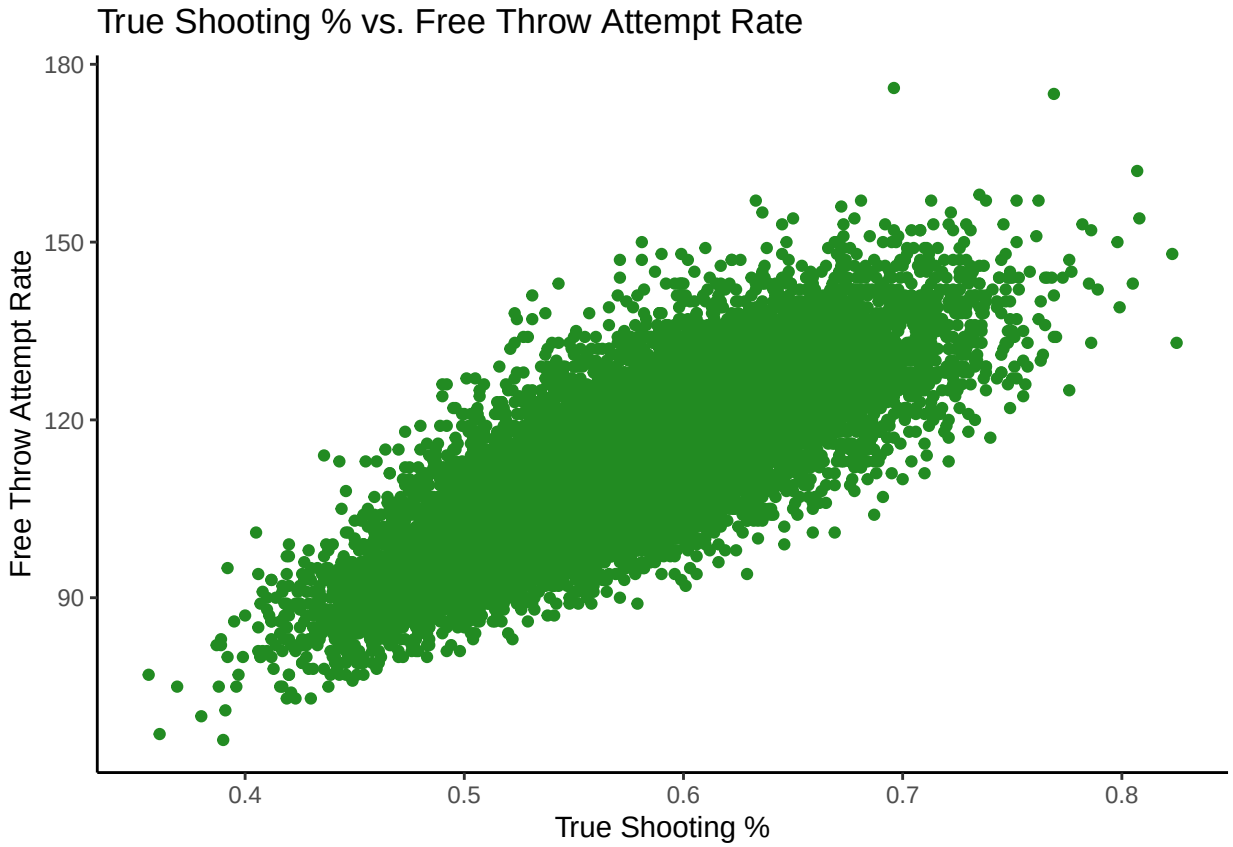

```
hist2plot <- ggplot(data=teamvisdata,
                    aes(x=ast_pct)) +
    geom_histogram(bins=30, color='black', fill='magenta3') +
    labs(
        title='Distribution of Shots Assisted On',
        x='% of Shots Assisted On',
        y='Count') +
    theme_bw() +
    scale_x_continuous(breaks = seq(from = 0, to = 100, by = 10))
hist2plot
```



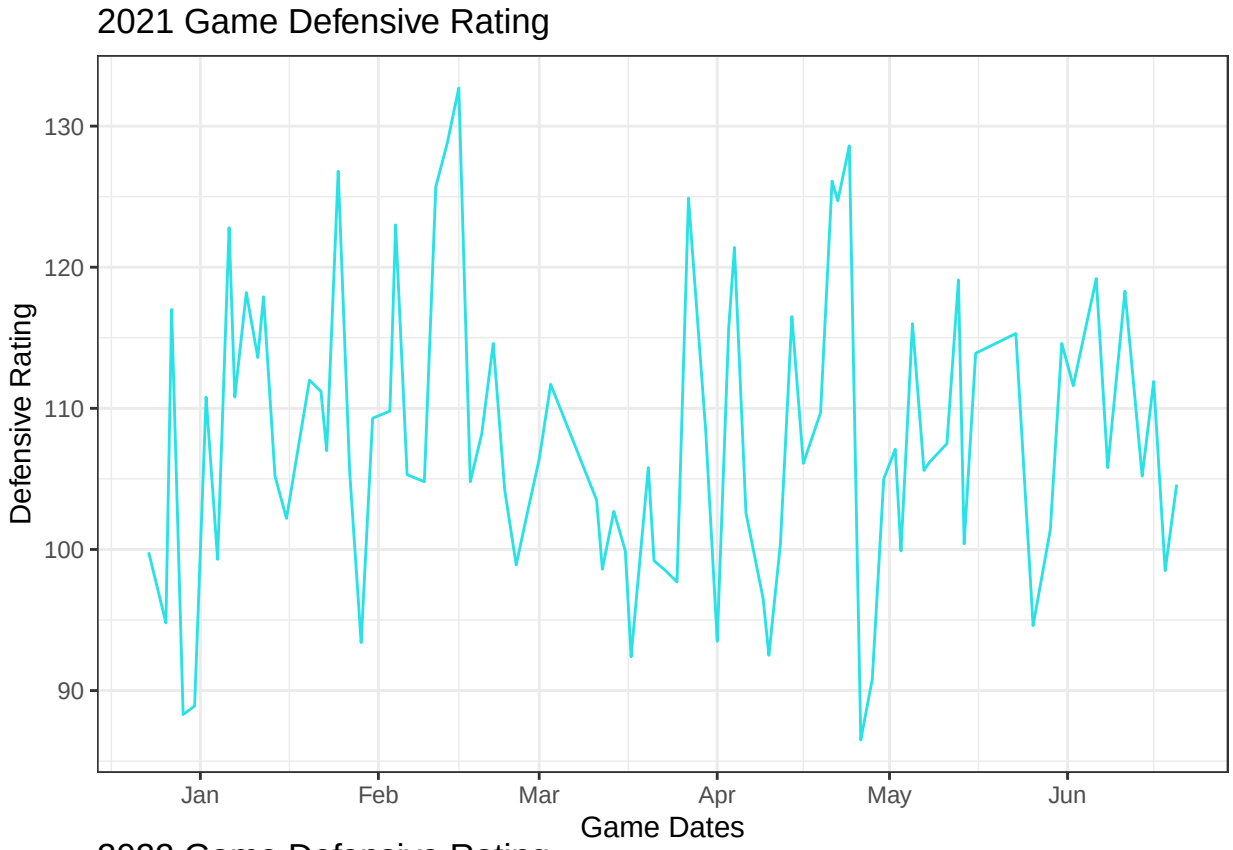

```
box1plot <- ggplot(data=teamvisdata,
                   aes(y=pts,
                       fill=twoleveloc)) +
    geom_boxplot() +
    facet_wrap(~season) +
    labs(
        title='Points Scored Faceted By Year (2020 to 2026)',
        x='Outcome of Game',
        y='Points Scored',
        fill='Game Outcome'
    ) +
    theme_bw()
box1plot
```

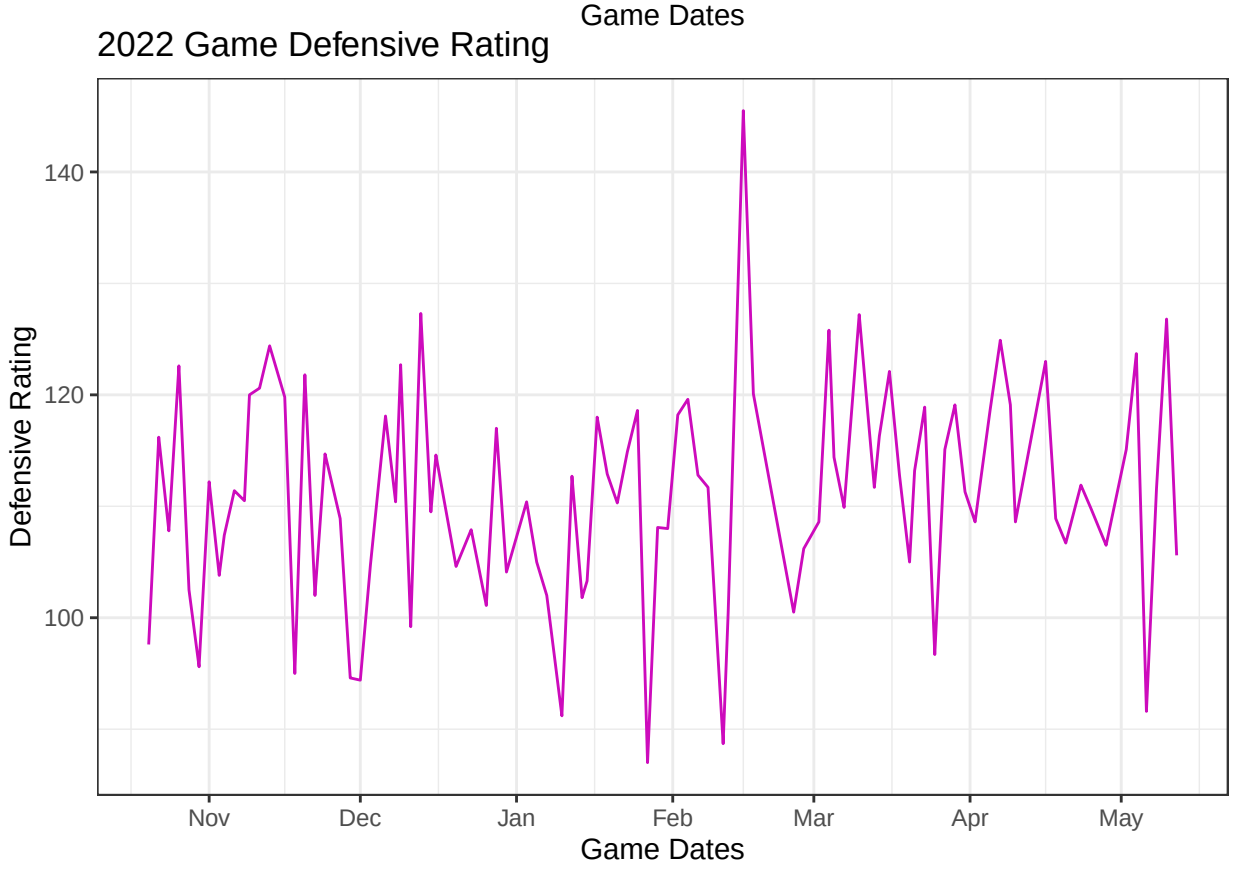


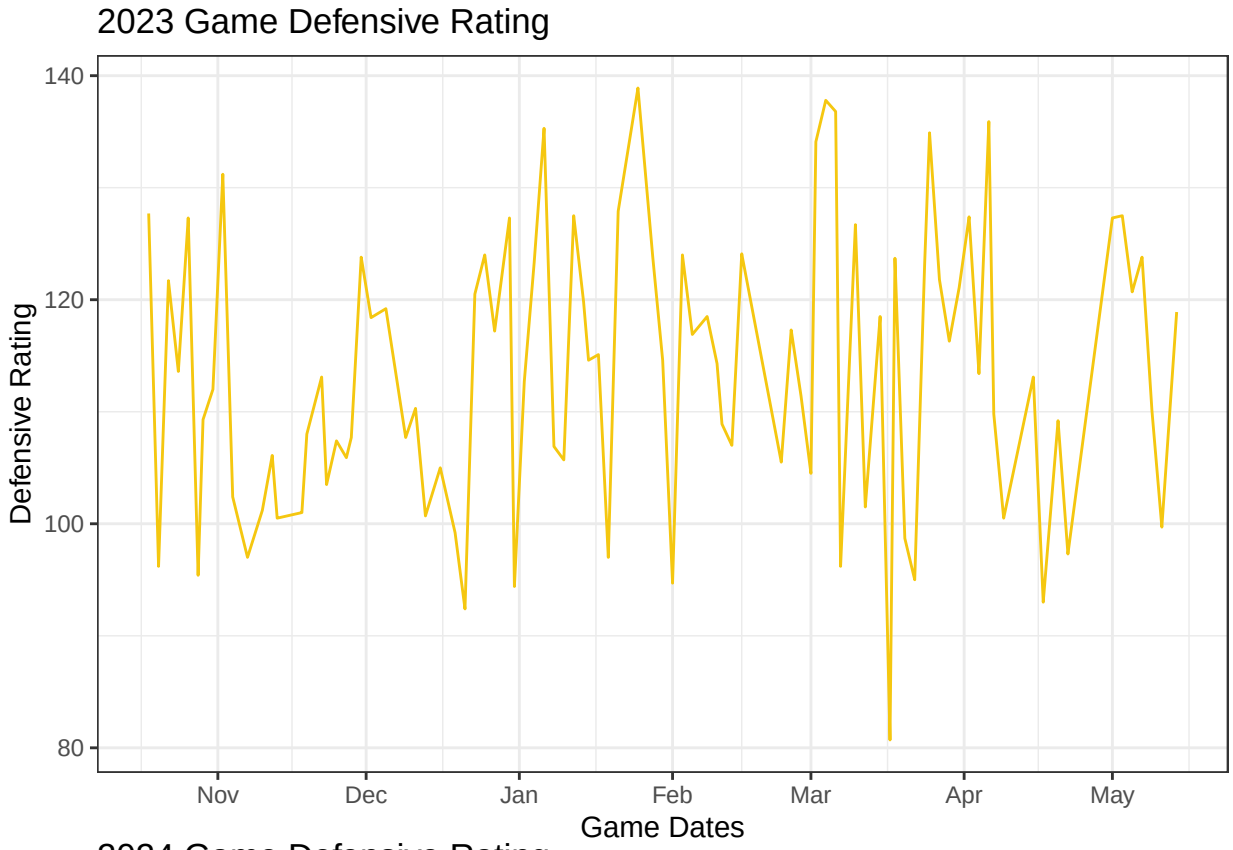

```
scat1plot <- ggplot(data=teamvisdata,
                    aes(x=ts_pct,
                        y=pts)) +
    geom_point(color='forestgreen') +
    labs(
        title='True Shooting % vs. Free Throw Attempt Rate',
        x='True Shooting %',
        y='Free Throw Attempt Rate') +
    theme_classic()
scat1plot
```

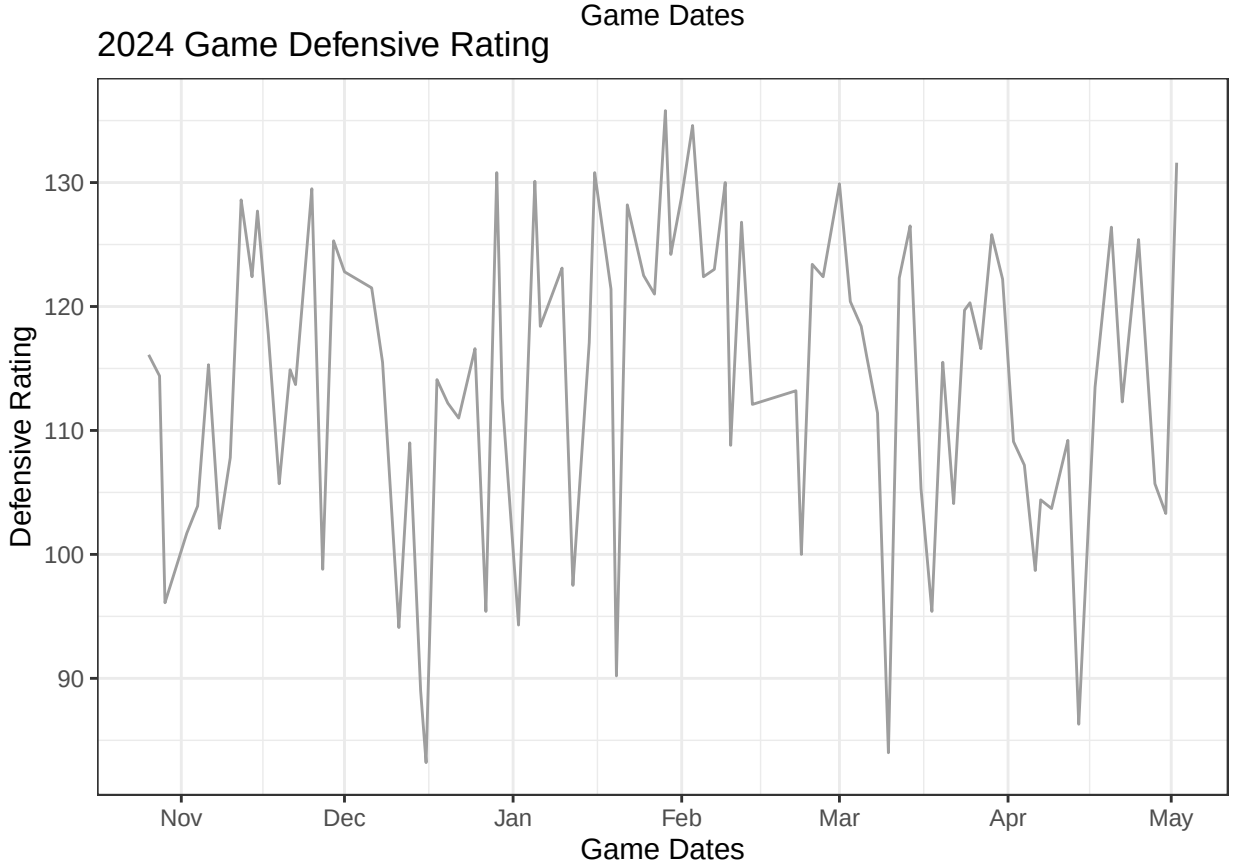


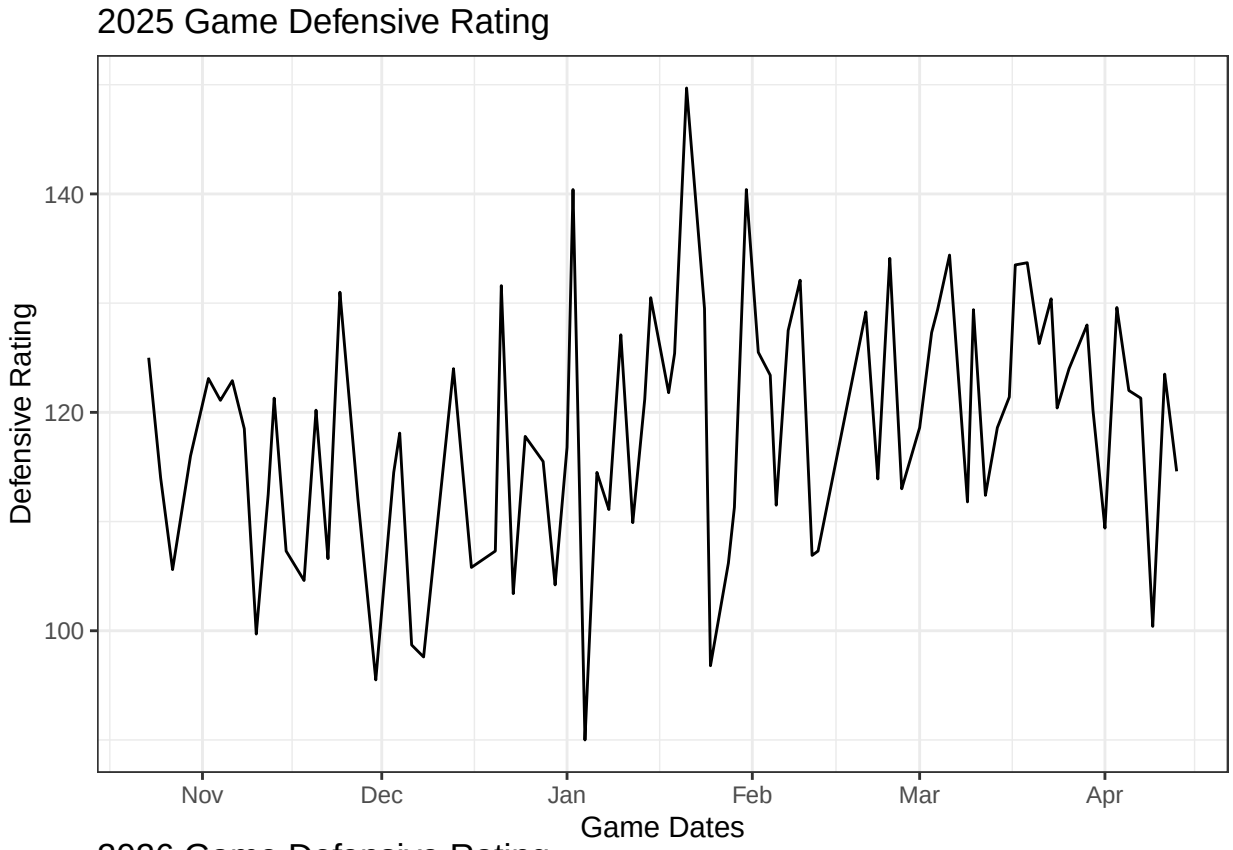

```
data_years <- sort(unique(teamvisdata$season))

for (data_year in data_years) {
    time1plot <- ggplot(data=teamvisdata %>%
                filter(team=='PHI',
                       season==data_year),
           aes(x=gamedate,
               y=defrating)) +
#         geom_point() +
        geom_line(color=data_year) +
        theme_bw() +
        scale_x_date(date_breaks='1 month', date_labels='%b') +
        labs(
            title=glue('{data_year} Game Defensive Rating'),
            x='Game Dates',
            y='Defensive Rating'
        )
    print(time1plot)
}
```

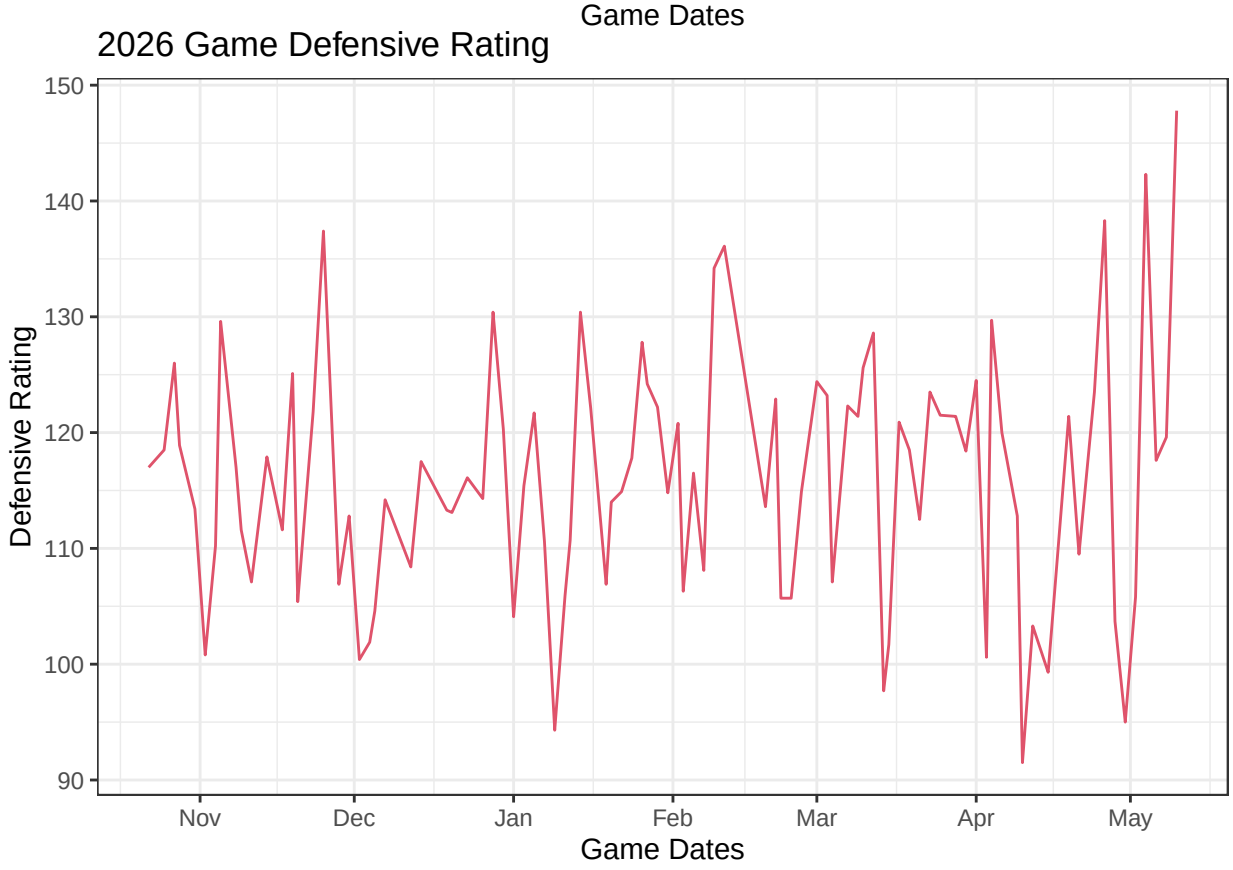

2021 Game Defensive Rating
Defensive Rating
130
120
110
100
90
Jan
Feb
Mar
Apr
May
Jun
Game Dates


2022 Game Defensive Rating
Defensive Rating
140
120
100
Nov
Dec
Jan
Feb
Mar
Apr
May
Game Dates

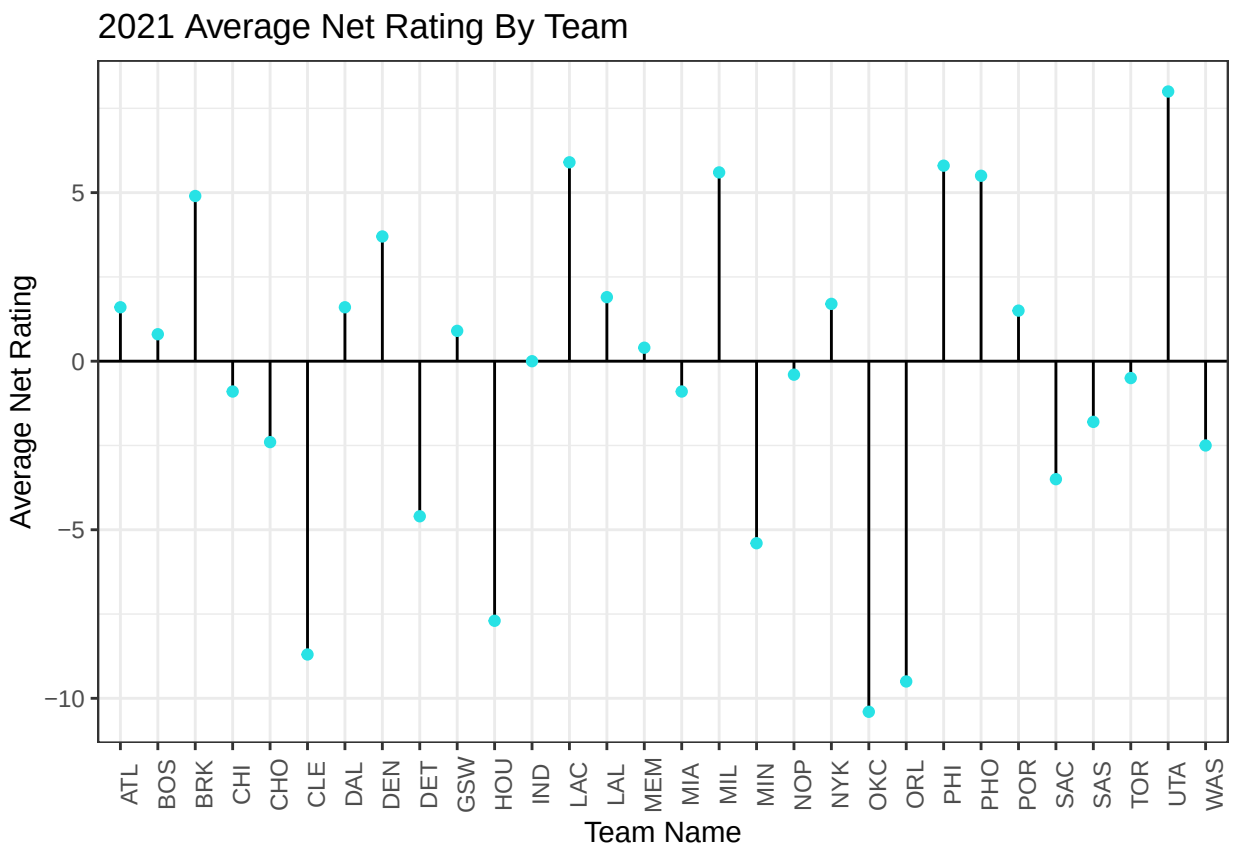

2023 Game Defensive Rating
Defensive Rating
140
120
100
80
Nov
Dec
Jan
Feb
Mar
Apr
May
Game Dates

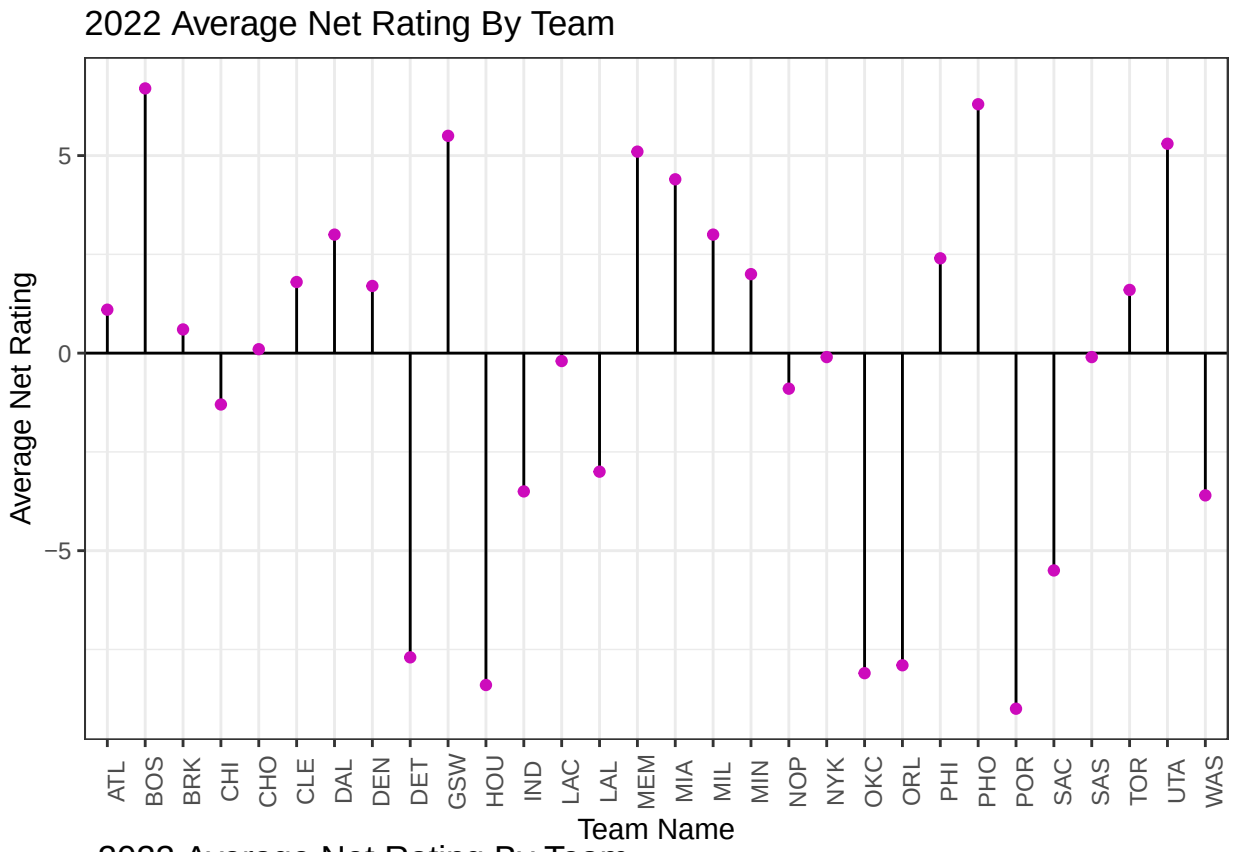

2024 Game Defensive Rating
Defensive Rating
130
120
110
100
90
Nov
Dec
Jan
Feb
Mar
Apr
May
Game Dates

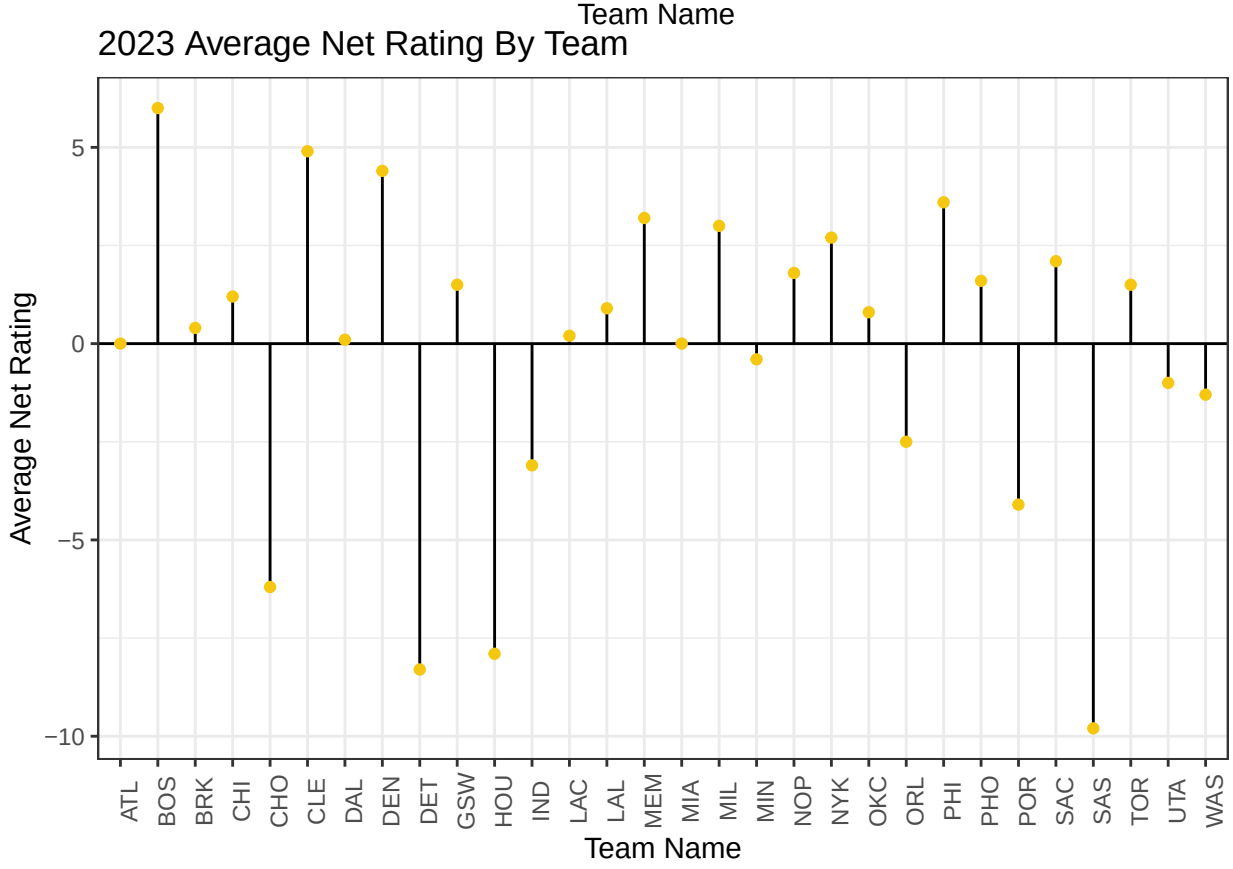

2025 Game Defensive Rating
Defensive Rating
140
120
100
Nov
Dec
Jan
Feb
Mar
Apr
Game Dates

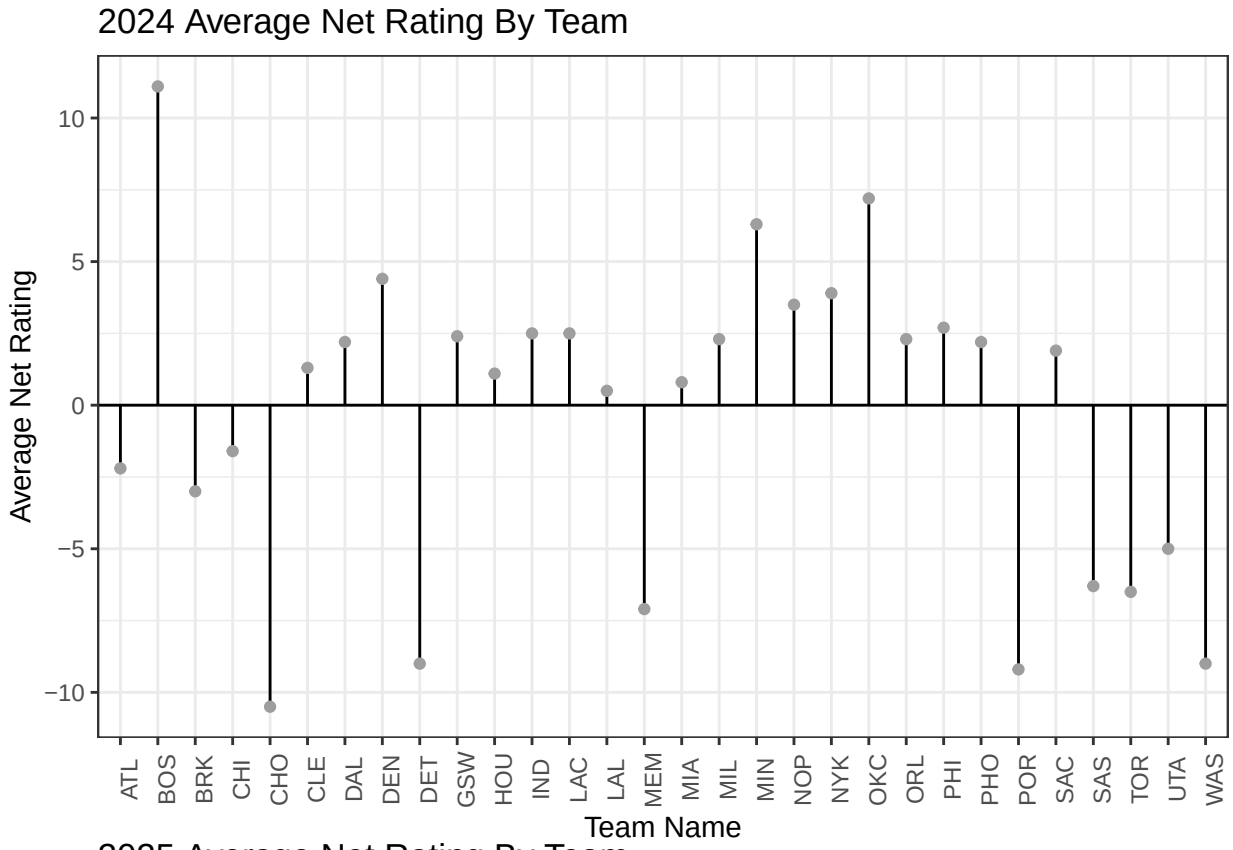

2026 Game Defensive Rating
Defensive Rating
150
140
130
120
110
100
90
Nov
Dec
Jan
Feb
Mar
Apr
May
Game Dates

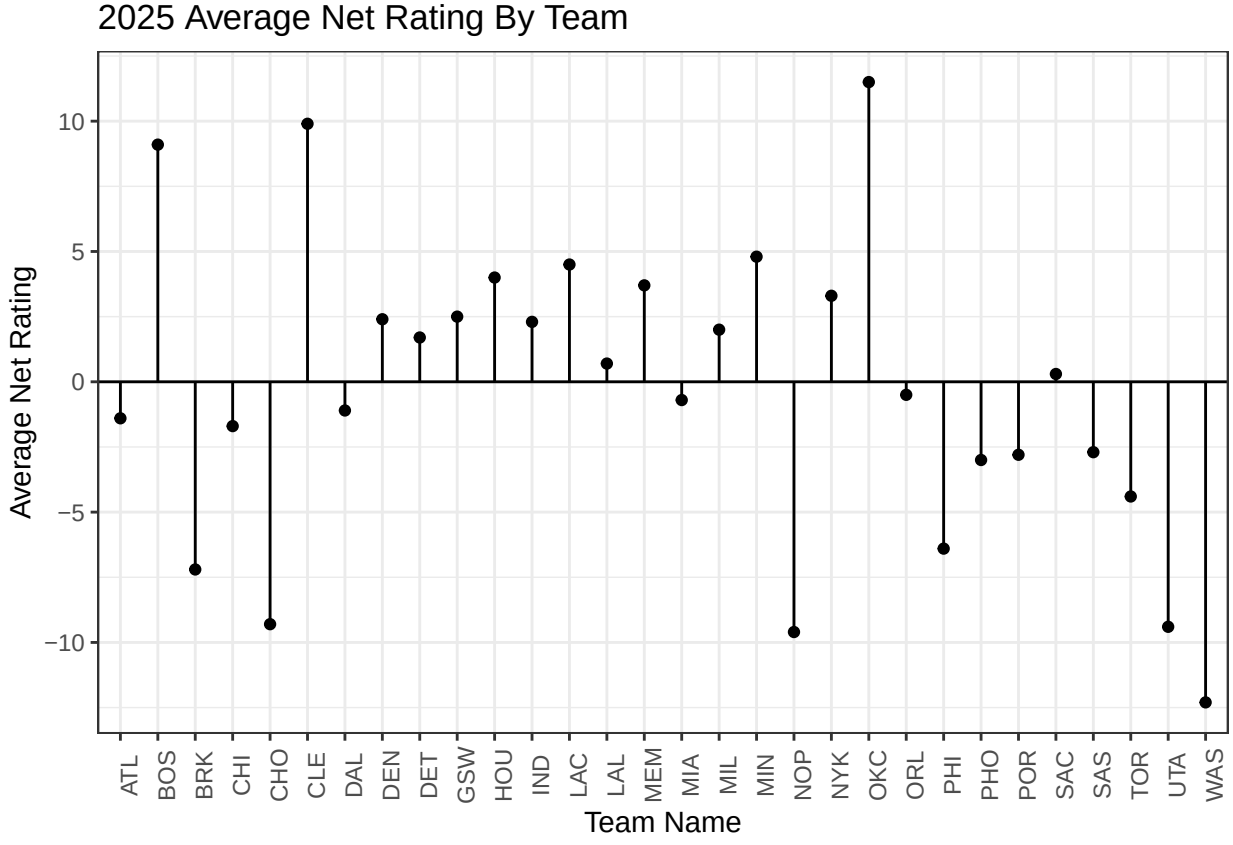
```
data_years <- sort(unique(teamvisdata$season))

for (data_year in data_years) {
    lol1plot <- ggplot(data=teamboxsummary %>%
                mutate(avg_netrating =
                            avg_offrating - avg_defrating) %>%
                filter(season==data_year),
            aes(x=team, y=avg_netrating)) +
        geom_segment(aes(
            x=team,
            xend=team,
            y=0,
            yend=avg_netrating), color='black') +
        theme_bw() +
        geom_hline(yintercept=0) +
        theme(axis.text.x = element_text(angle = 90)) +
        geom_point(color=data_year) +
        labs(
            title=glue('{data_year} Average Net Rating By Team'),
            x='Team Name',
            y='Average Net Rating'
        )
    print(lol1plot)
}
```

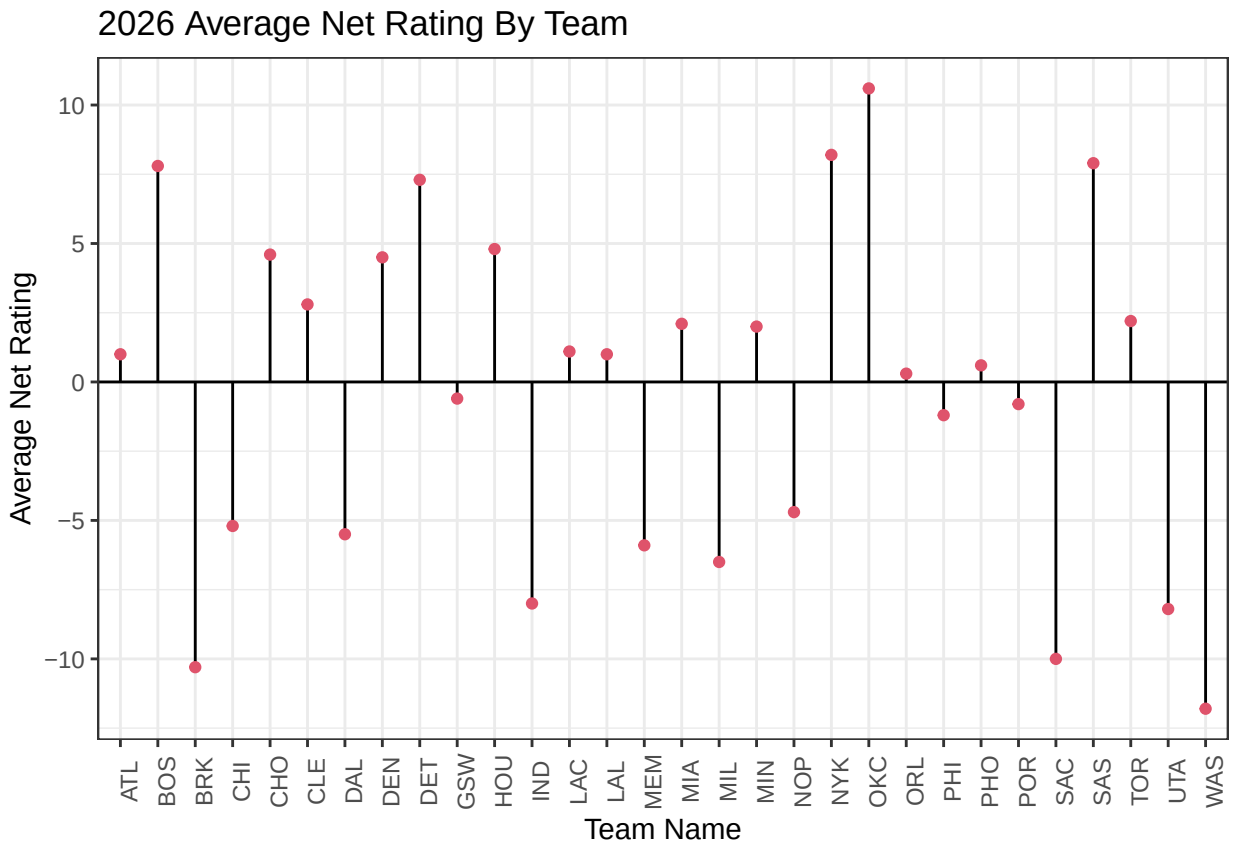



2022 Average Net Rating By Team
Average Net Rating
5
0
−5
ATL
BOS
BRK
CHI
CHO
CLE
DAL
DEN
DET
GSW
HOU
IND
LAC
LAL
MEM
MIA
MIL
MIN
NOP
NYK
OKC
ORL
PHI
PHO
POR
SAC
SAS
TOR
UTA
WAS
Team Name


2023 Average Net Rating By Team
Average Net Rating
5
0
−5
−10
ATL
BOS
BRK
CHI
CHO
CLE
DAL
DEN
DET
GSW
HOU
IND
LAC
LAL
MEM
MIA
MIL
MIN
NOP
NYK
OKC
ORL
PHI
PHO
POR
SAC
SAS
TOR
UTA
WAS
Team Name


2024 Average Net Rating By Team
Average Net Rating
10
5
0
−5
−10
ATL
BOS
BRK
CHI
CHO
CLE
DAL
DEN
DET
GSW
HOU
IND
LAC
LAL
MEM
MIA
MIL
MIN
NOP
NYK
OKC
ORL
PHI
PHO
POR
SAC
SAS
TOR
UTA
WAS
Team Name


2025 Average Net Rating By Team
Average Net Rating
10
5
0
−5
−10
ATL
BOS
BRK
CHI
CHO
CLE
DAL
DEN
DET
GSW
HOU
IND
LAC
LAL
MEM
MIA
MIL
MIN
NOP
NYK
OKC
ORL
PHI
PHO
POR
SAC
SAS
TOR
UTA
WAS
Team Name


2026 Average Net Rating By Team
Average Net Rating
10
5
0
−5
−10
ATL
BOS
BRK
CHI
CHO
CLE
DAL
DEN
DET
GSW
HOU
IND
LAC
LAL
MEM
MIA
MIL
MIN
NOP
NYK
OKC
ORL
PHI
PHO
POR
SAC
SAS
TOR
UTA
WAS
Team Name